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# Practical No. 1

AIM: Programming Resources Android Resources: (Color, Theme, String, Drawable, Dimension, Image)

## CODE:

### Color.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="white">#FB0707</color>
</resources>
```

### String.xml

```
<resources>
    <string name="app_name">pract2</string>
    <string name="app_name">String</string>
</resources>
```

### Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/Colg"
    tools:context=".MainActivity">
</androidx.constraintlayout.widget.ConstraintLayout>
```

### MainActivity.kt

```
package com.example.pract2
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }
}
```

**theme.xml:**

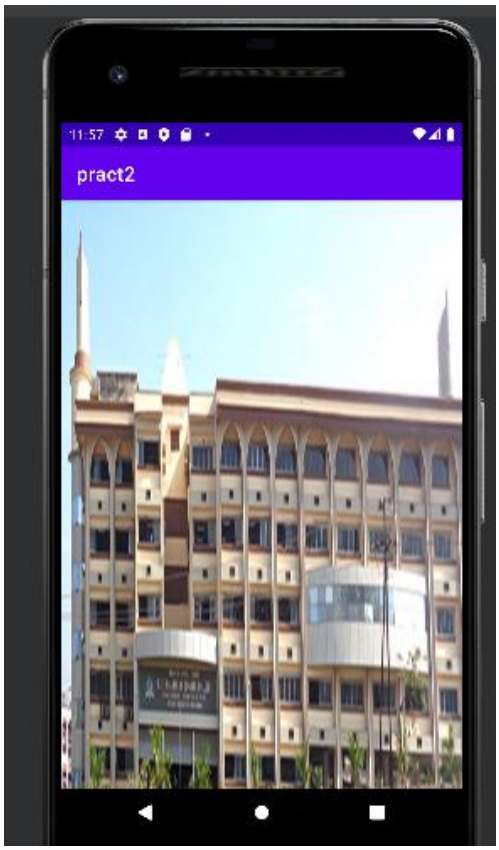
```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Menu" parent="Theme.AppCompact.DayNight.DarkActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.Menu" parent="Base.Theme.Menu" />
</resources>
```

**night/theme.xml:**

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Menu" parent="Theme.AppCompact.DayNight.DarkActionBar">
        <!-- Customize your dark theme here. -->
        <!-- <item name="colorPrimary">@color/my_dark_primary</item> -->
    </style>
</resources>
```

**<u>OUTPUT:</u>**



# Practical No. 2:

AIM: Programming Activities and fragments

## CODE:

### Activity_main.xml

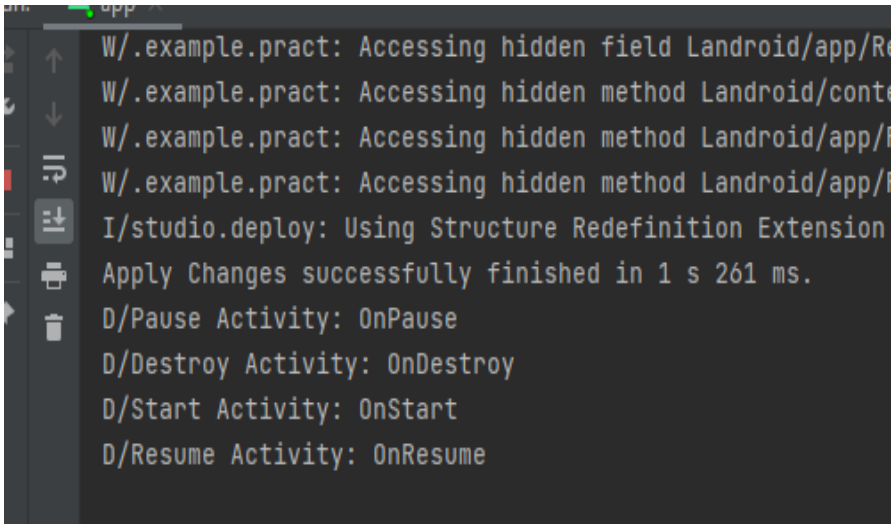
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <Button
        android:id="@+id/MainActivity"
        android:layout_width="262dp"
        android:layout_height="81dp"
        android:text="CLick Me"
        tools:layout_editor_absoluteX="0dp"
        tools:layout_editor_absoluteY="2dp">
    </Button>
</androidx.constraintlayout.widget.ConstraintLayout>
```

### MainActivity.kt

```
package com.example.lifecycleactivity
import android.util.Log
import android.widget.Toast
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    private val TAG = "MainActivity"
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        Log.d(TAG, "In Oncreate")
        }
```

```
    override fun onStart() {
        super.onStart()
        Toast.makeText(this,"onStart Activity", Toast.LENGTH_SHORT).show()
        Log.d(TAG,"In onStart")
    }
    override fun onResume() {
        super.onResume()
        Toast.makeText(this,"onResume Activity", Toast.LENGTH_SHORT).show()
        Log.d(TAG,"In onResume")
    }
    override fun onPause() {
        super.onPause()
        Toast.makeText(this,"onPause Activity", Toast.LENGTH_SHORT).show()
        Log.d(TAG,"In onPause")
    }
    override fun onStop() {
        super.onStop()
        Toast.makeText(this,"onStop Activity", Toast.LENGTH_SHORT).show()
        Log.d(TAG,"In onStop")
    }
    override fun onRestart() {
        super.onRestart()
        Toast.makeText(this,"onRestart Activity", Toast.LENGTH_SHORT).show()
        Log.d(TAG,"In onRestart")
    }
    override fun onDestroy() {
        super.onDestroy()
        Toast.makeText(this,"onDestroy Activity", Toast.LENGTH_SHORT).show()
        Log.d(TAG,"In onDestroy")
    }
}
```

**OUTPUT:**



# Practical No. 3:

AIM: Programs related to different Layouts: Linear, Relative, Absolute, Table, Frame, List View, Grid View.

## CODE:

## Linear Layout & Frame Layout:

### Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <FrameLayout
      android:layout_width="wrap_content"
      android:layout_height="wrap_content">
      <TextView
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Hello World!"
         app:layout_constraintBottom_toBottomOf="parent"
         app:layout_constraintEnd_toEndOf="parent"
         app:layout_constraintStart_toStartOf="parent"
         app:layout_constraintTop_toTopOf="parent" />
      <ImageView
         android:id="@+id/imageView3"
         android:layout_width="481dp"
         android:layout_height="match_parent"
         android:layout_weight="10"
         app:srcCompat="@drawable/images" />
   </FrameLayout>
</LinearLayout>
```

### MainActivity.kt

```
package com.example.practicepractical
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
```

```
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
    }
}
```

**OUTPUT:**

**GridLayout:**

**Activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="0"
    android:layout_column="0"
    android:text="1" />
<Button
    android:id="@+id/button2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="0"
    android:layout_column="1"
    android:text="2" />
<Button
    android:id="@+id/button3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="0"
    android:layout_column="2"
    android:text="3" />
<Button
    android:id="@+id/button4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="1"
    android:layout_column="0"
    android:text="4" />
<Button
    android:id="@+id/button5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="1"
    android:layout_column="1"
    android:text="5" />
<Button
    android:id="@+id/button6"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="1"
```

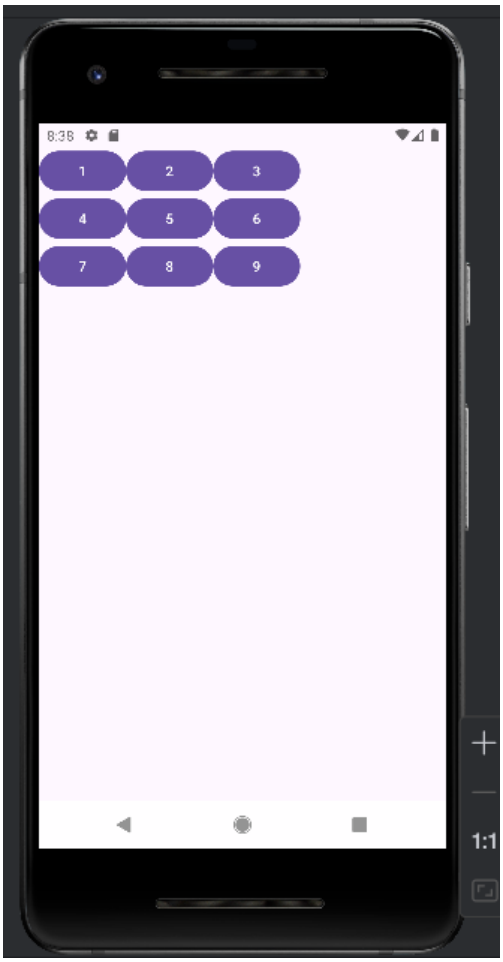
```
        android:layout_column="2"
        android:text="6" />
    <Button
        android:id="@+id/button7"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_row="2"
        android:layout_column="0"
       android:text="7" />
    <Button
        android:id="@+id/button8"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_row="2"
        android:layout_column="1"
        android:text="8" />
    <Button
        android:id="@+id/button9"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_row="2"
        android:layout_column="2"
        android:text="9" />
</GridLayout>
```

**MainActivity.kt**

```
package com.example.prac3
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
```

```
    }
}
```

**OUTPUT:**




## Relative Layout & Table Layout:

### Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TableLayout
        android:layout_width="663dp"
        android:layout_height="1230dp"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">
```

```
    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent" >
        <Button
            android:id="@+id/button5"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="1" />
        <Button
            android:id="@+id/button"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="2" />
        <Button
            android:id="@+id/button4"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="3" />
    </TableRow>
    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent" >
        <Button
            android:id="@+id/button6"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="4" />
        <Button
            android:id="@+id/button2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="5" />
        <Button
            android:id="@+id/button3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="6" />
    </TableRow>
  </TableLayout>
</RelativeLayout>
```

**MainActivity.kt**

```
package com.example.myapplication
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
    }
}
```

**OUTPUT:**

## List View:

### Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ListView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:entries="@array/insertlist"
    tools:context=".MainActivity">
</ListView>
```

### MainActivity.kt

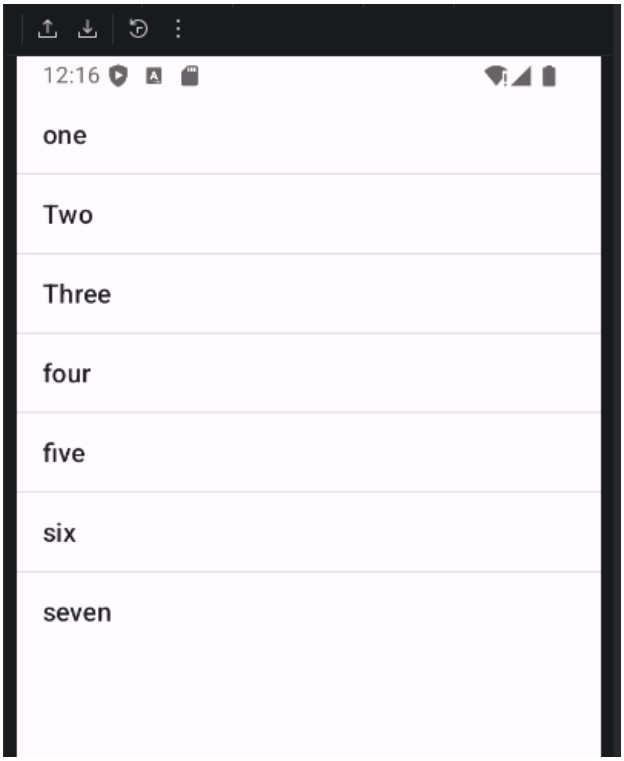
```
package com.example.myapplication
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
    }
}
```

## String.xml:

```
<resources>
    <string name="app_name">My Application</string>
    <array name="insertlist">
        <Item>one</Item>
        <Item>Two</Item>
        <Item>Three</Item>
        <Item>four</Item>
```

```
        <Item>five</Item>
        <Item>six</Item>
        <Item>seven</Item>
    </array>
</resources>
```

**OUTPUT:**




**Absolute Layout:**

**Activity_main.xml:**

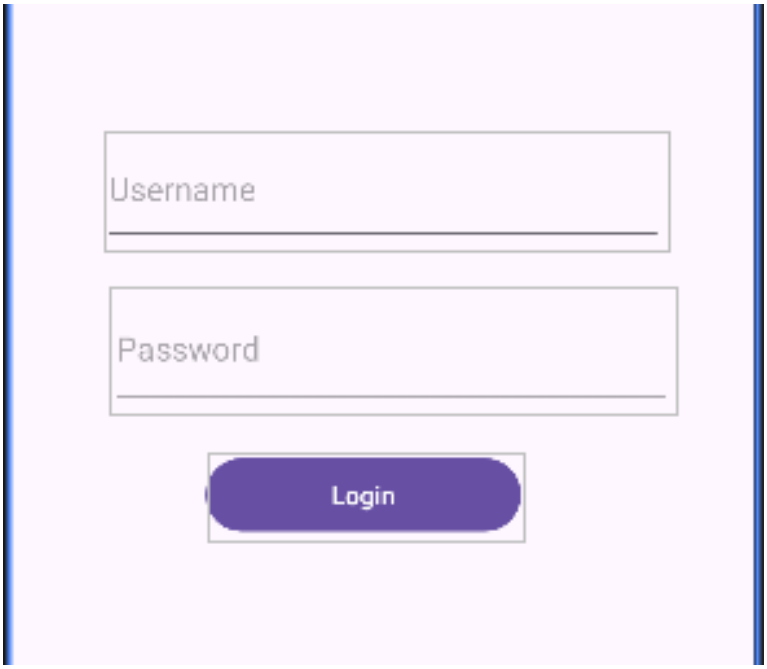
```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <EditText
        android:id="@+id/editTextText"
        android:layout_width="310dp"
        android:layout_height="65dp"
        android:layout_x="52dp"
        android:layout_y="152dp"
```

```
        android:ems="10"
        android:hint="Username"
        android:inputType="text" />
    <EditText
        android:id="@+id/editTextTextPassword"
        android:layout_width="311dp"
        android:layout_height="70dp"
        android:layout_x="56dp"
        android:layout_y="237dp"
        android:ems="10"
        android:hint="Password"
        android:inputType="textPassword" />
    <Button
        android:id="@+id/button"
        android:layout_width="174dp"
        android:layout_height="wrap_content"
        android:layout_x="109dp"
        android:layout_y="328dp"
        android:text="Login" />
</AbsoluteLayout>
```

**MainActivity.kt:**

```
package com.example.absolutelayout
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
    }
}
```

**OUTPUT:**



# Practical No. 4:

AIM: Programming UI elements: AppBar, Fragments.

## A) AppBar

**CODE:**

**activity_main.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

**MainActivity.kt:**

```
package com.example.Appbar
import android.os.Bundle
import android.view.Menu
import android.view.MenuItem
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
```

```
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
    }
    override fun onCreateOptionsMenu(menu: Menu?): Boolean {
        menuInflater.inflate(R.menu.menu, menu)
        return true
    }
    override fun onOptionsItemSelected(item: MenuItem): Boolean {
        when (item.itemId) {
            R.id.search -> {
                Toast.makeText(this, "Search menu selected", Toast.LENGTH_LONG).show()
                return true
            }
            R.id.aboutus -> {
                Toast.makeText(this, "About us Selected", Toast.LENGTH_LONG).show()
                return true
            }
            R.id.contactus -> {
                Toast.makeText(this, "Contact menu selected", Toast.LENGTH_LONG).show()
                return true
            }
            else -> return super.onOptionsItemSelected(item)
        }
    }
}
```

**Menu.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
<item android:title="search"
    android:id="@+id/search"
    android:icon="@color/white"
    app:showAsAction="always">
    </item>
    <item android:id="@+id/aboutus"
        android:title="About us">
        </item>
    <item
        android:id="@+id/contactus"
        android:title="Contact us">
        </item>
```

```
</menu>
```

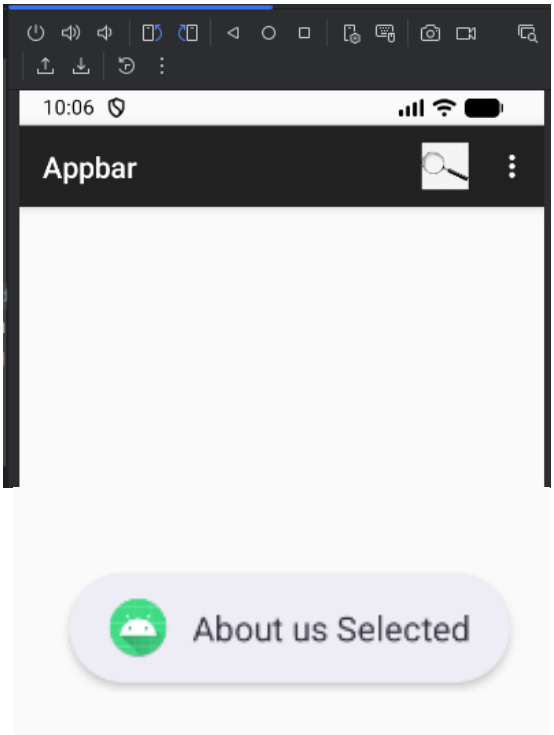
**theme.xml:**

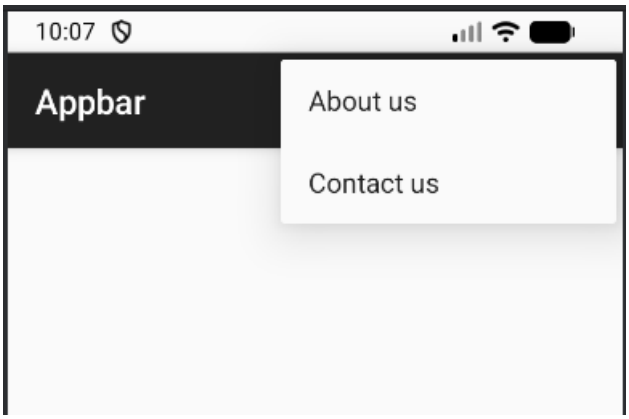
```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Menu" parent="Theme.AppCompact.DayNight.DarkActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>
    <style name="Theme.Menu" parent="Base.Theme.Menu" />
</resources>
```

**night/theme.xml:**

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Menu" parent="Theme.AppCompact.DayNight.DarkActionBar">
        <!-- Customize your dark theme here. -->
        <!-- <item name="colorPrimary">@color/my_dark_primary</item> -->
    </style>
</resources>
```

**<u>OUTPUT:</u>**



## B) Fragments

**CODE:**

**Activity_main.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <androidx.fragment.app.FragmentContainerView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/fragmentContainer"/>
    <Button
        android:layout_width="150dp"
        android:layout_height="60dp"
        android:id="@+id/fragment1btn"
        android:layout_alignParentBottom="true"
        android:layout_marginBottom="50dp"
        android:layout_marginStart="30dp"
        android:text="Fragment1"/>
    <Button
        android:layout_width="150dp"
        android:layout_height="60dp"
        android:id="@+id/fragment2btn"
        android:layout_alignParentBottom="true"
        android:layout_alignParentRight="true"
        android:layout_marginBottom="50dp"
        android:layout_marginStart="30dp"
        android:text="Fragment2"/>
</RelativeLayout>
```

**MainActivity.kt:**

```
package com.example.fragments
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.fragment.app.Fragment
import androidx.fragment.app.FragmentManager
```

```
import com.example.fragments.databinding.ActivityMainBinding
class MainActivity : AppCompatActivity() {
   private lateinit var fragmentManager: FragmentManager
   private lateinit var binding:ActivityMainBinding
      override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      binding = ActivityMainBinding.inflate(layoutInflater)
      setContentView(binding.root)

      binding.fragment1btn.setOnClickListener {
         goToFragment(Fragment1())
      }
      binding.fragment2btn.setOnClickListener {
         goToFragment(Fragment2())
      }
   }
   private fun goToFragment(fragment: Fragment)
   {
      fragmentManager= supportFragmentManager
      fragmentManager.beginTransaction().replace(R.id.fragmentContainer,fragment).commit()
   }
}
```

**Fragment_1.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:background="@color/black">
   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:gravity="center"
      android:textSize="50dp"
      android:text="fragment1" />
</RelativeLayout>
```

**Fragment1.kt:**

```
package com.example.fragments
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
```

```
import android.view.View
import android.view.ViewGroup
class Fragment1 : Fragment() {
    override fun onCreateView(
        inflater: LayoutInflater, container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View? {
        // Inflate the layout for this fragment
        return inflater.inflate(R.layout.activity_fragment1, container, false)
    }
}
```

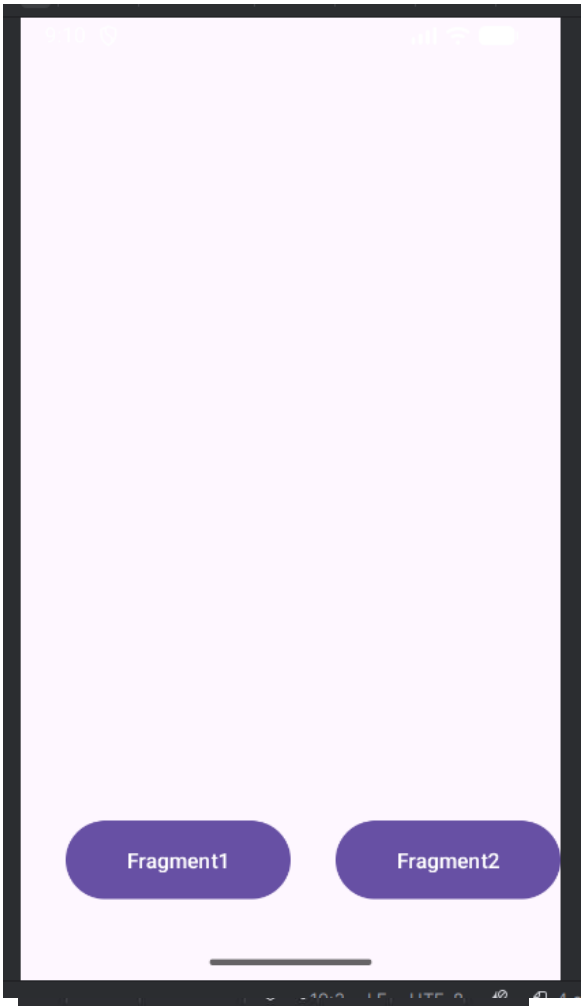
**Fragment_2.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/pink">
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textSize="50dp"
        android:text="fragment2" />
</RelativeLayout>
```

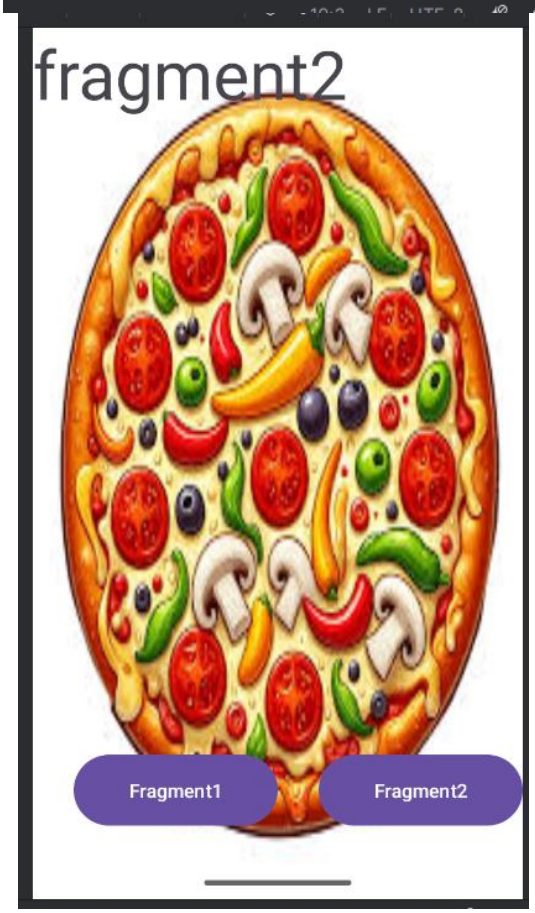
**Fragment2.kt:**

```
package com.example.fragments
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
class Fragment2 : Fragment() {
    override fun onCreateView(
        inflater: LayoutInflater, container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View? {
        // Inflate the layout for this fragment
        return inflater.inflate(R.layout.activity_fragment2, container, false)
    }
}
```

**OUTPUT:**


Fragment1
Fragment2

fragment1
Fragment1
Fragment2


fragment2
Fragment1
Fragment2

# Practical No. 5:

AIM: Programs on Intent

## CODE:

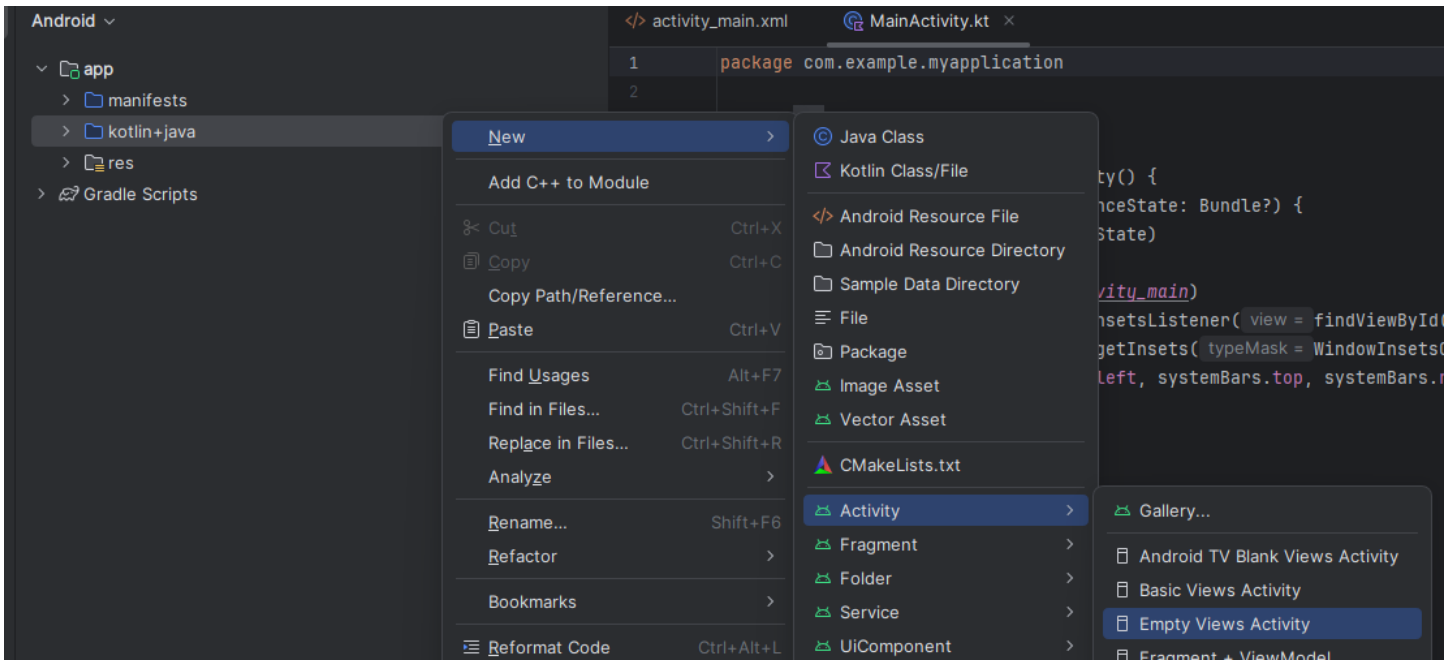
### activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <EditText
      android:id="@+id/mytext"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:ems="10"
      android:hint="Enter your message"
      android:inputType="textPersonName"
      android:text="Name" />
   <Button
      android:id="@+id/send"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_gravity="center_horizontal"
      android:layout_margin="20dp"
      android:text="send"
      tools:layout_editor_absoluteX="161dp"
      tools:layout_editor_absoluteY="181dp" />
   </RelativeLayout>
```

### MainActivity.kt:

```
package com.example.pract6
import android.content.Intent
import android.os.Bundle
import android.view.KeyEvent
import android.view.View
import android.widget.Button
import android.widget.EditText
import androidx.activity.enableEdgeToEdge
```

```
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)
      val message= findViewById<EditText>(R.id.mytext)
      val button= findViewById<Button>(R.id.send)
      button.setOnKeyListener(object : View.OnKeyListener
      {
         fun onClick(v: View?)
         {
            intent= Intent(applicationContext, SecondActivity::class.java)
            intent.putExtra("msg",message.text.toString())
            startActivity(intent)
         }
         override fun onKey(
            v: View?,
            keyCode: Int,
            event: KeyEvent?
         ): Boolean {
            TODO("Not yet implemented")
         }
      })
   }
   }
```



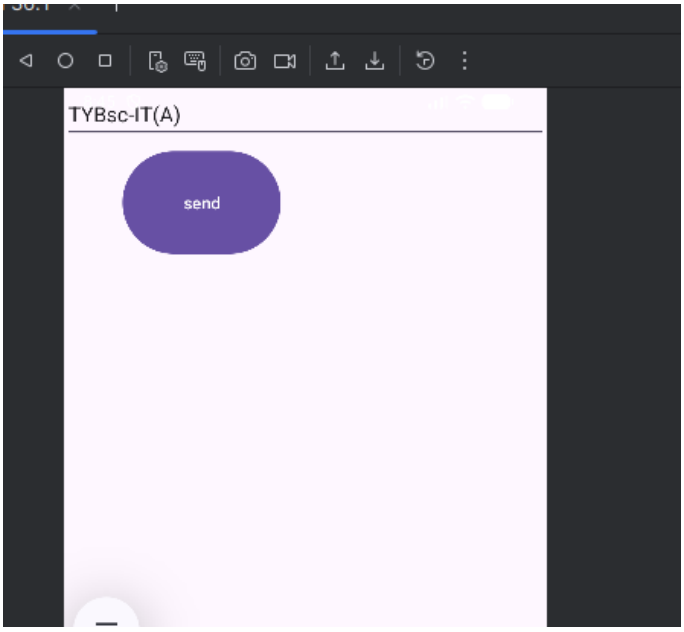
**activity_second.xml:**

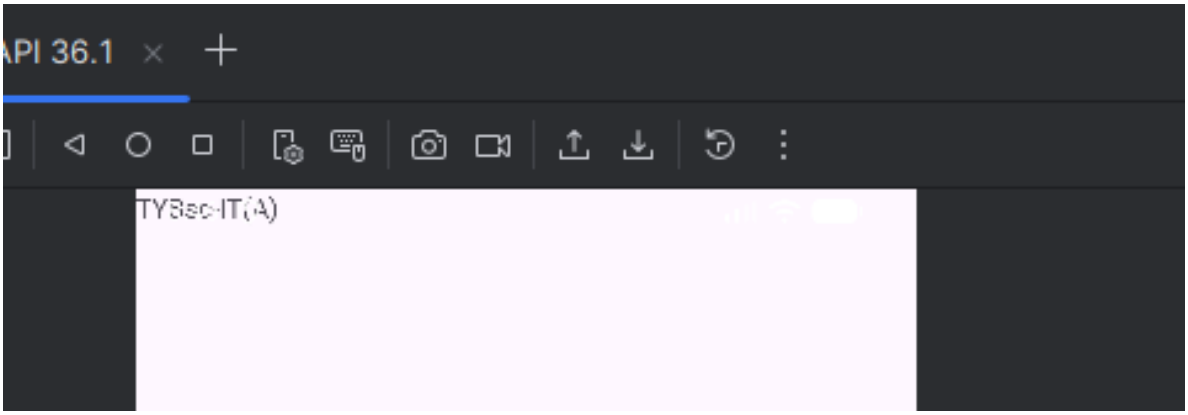
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SecondActivity">
   <TextView
        android:id="@+id/display"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="TextView" />
</LinearLayout>
```

**SecondActivity.kt:**

```
package com.example.pract6
import android.os.Bundle
import android.widget.TextView
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class SecondActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_second)
        val text_display = findViewById<TextView>(R.id.display)
        text_display.setText(intent.extras!!.getString("msg"))
    }
}
```

OUTPUT:



# Practical No. 6

AIM: Programming menus, dialog, dialog fragments

## A) MENU

## CODE:

### Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello World!"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintLeft_toLeftOf="parent"
      app:layout_constraintRight_toRightOf="parent"
      app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

### Menu.xml:

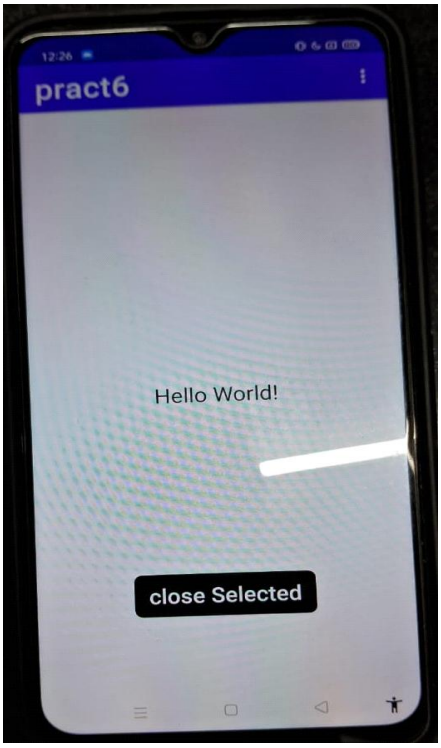
```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
<item  android:title="open"
   android:id="@+id/open"/>
   <item  android:title="close"
      android:id="@+id/close"/>
   <item  android:title="save"
      android:id="@+id/save"/>
</menu>
```

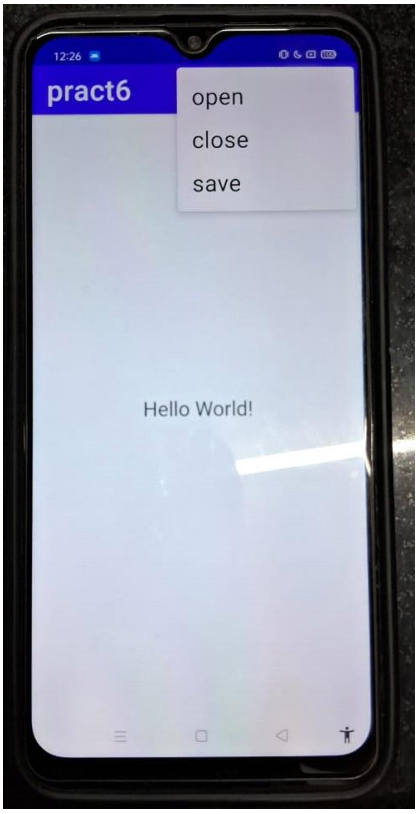
### MainActivity.kt:

```
package com.example.pract6
import androidx.annotation.NonNull;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
   }
   @Override
   public boolean onCreateOptionsMenu(Menu menu) {
      getMenuInflater().inflate(R.menu.menus,menu);
      return true;
   }
   @Override
   public boolean onOptionsItemSelected(@NonNull MenuItem item) {
      int id=item.getItemId();
      if(id==R.id.open){
         Toast.makeText(this, "Open Selected", Toast.LENGTH_SHORT).show();
      }
      else if(id==R.id.close) {
         Toast.makeText(this, "close Selected", Toast.LENGTH_SHORT).show();
      }
      else if(id==R.id.save) {
         Toast.makeText(this, "save Selected", Toast.LENGTH_SHORT).show();
      }
      else{
         return super.onOptionsItemSelected(item);
      }
    return true;
   }
}
```

# OUTPUT:


12:26
pract6
Hello World!
close Selected


12:26
pract6
open
close
save
Hello World!

## B) Dialog & Dialog fragments:

**CODE:**

### activity_main.xml:

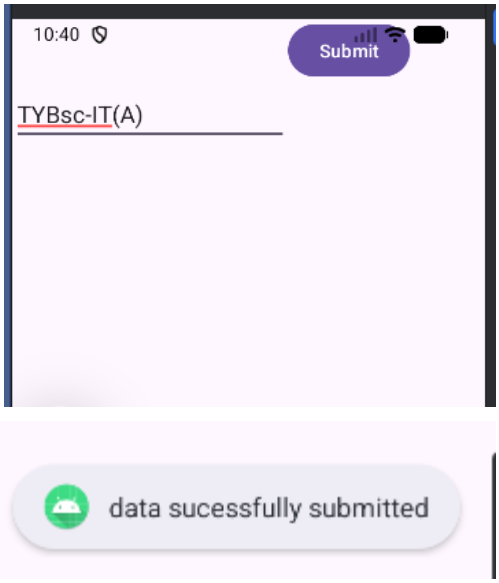
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <EditText
      android:id="@+id/editText1"
      android:layout_marginTop="50dp"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:ems="10"
      android:inputType="text"
      android:text="Name"
      tools:layout_editor_absoluteX = "86dp"
      tools:layout_editor_absoluteY = "33dp"/>
   <Button
      android:id="@+id/alert"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginTop="150dp"
      android:layout_gravity=""
      android:text="Submit"
      tools:layout_editor_absoluteX = "124dp"
      tools:layout_editor_absoluteY = "160dp"/>
</LinearLayout>
```

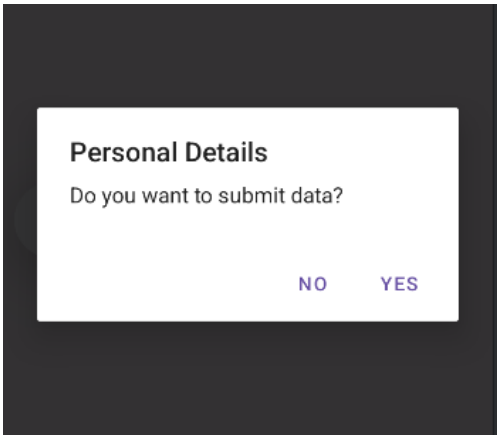
### MainActivity.kt:

```
package com.example.alert
import android.app.AlertDialog
import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
```

```
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      enableEdgeToEdge()
      setContentView(R.layout.activity_main)
      val submit = findViewById<Button>(R.id.alert)
      submit.setOnClickListener {
         showAlertDialog()
      }
   }
   private fun showAlertDialog() {
      val builder = AlertDialog.Builder(this)
      builder.setTitle("Personal Details")
         .setMessage("Do you want to submit data?")
         .setPositiveButton("Yes") { dialog, which ->
            Toast.makeText(this,"data sucessfully submitted", Toast.LENGTH_LONG).show()
         }
         .setNegativeButton("No"){ dialog, which -> dialog.dismiss()
         }
      val alertDialog: AlertDialog = builder.create()
      alertDialog.show()
   }
   }
```

**OUTPUT:**



# Practical No. 7

AIM: Programming Security and permissions

## CODE:

### activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

### MainActivity.kt:

```
package com.example.pract6
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.content.ContextCompat
import android.content.pm.PackageManager
import android.util.Log
import androidx.core.app.ActivityCompat
import androidx.appcompat.app.AlertDialog
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
    private val TAG = "permission denied"
    val REQUEST_RECORD_CODE=1
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
```

```
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        setpermission()
    }
    private fun setpermission(){
        val permission = ContextCompat.checkSelfPermission(
            this,android.Manifest.permission.RECORD_AUDIO)
        if (permission != PackageManager.PERMISSION_GRANTED){
            Log.d(TAG,"permission denied")
        }
        if (ActivityCompat.shouldShowRequestPermissionRationale(
            this,android.Manifest.permission.RECORD_AUDIO))
        {
            val builder = AlertDialog.Builder(this)
            builder.setMessage("permission to access to microphone")
            builder.setTitle("permission required")
            builder.setPositiveButton("ok")
            {
                dialog, which ->
                Log.d(TAG,"clicked")
                makeRequest()
            }
            val dialog=builder.create()
            dialog.show()
        }
        else{
            makeRequest()

        }
    }
    private fun makeRequest()
    {
ActivityCompat.requestPermissions(this,arrayOf(android.Manifest.permission.RECORD_AUD
IO),REQUEST_RECORD_CODE)
    }
    override fun onRequestPermissionsResult(
        requestCode: Int,
        permissions: Array<out String?>,
        grantResults: IntArray,
        deviceId: Int
    ) {
```

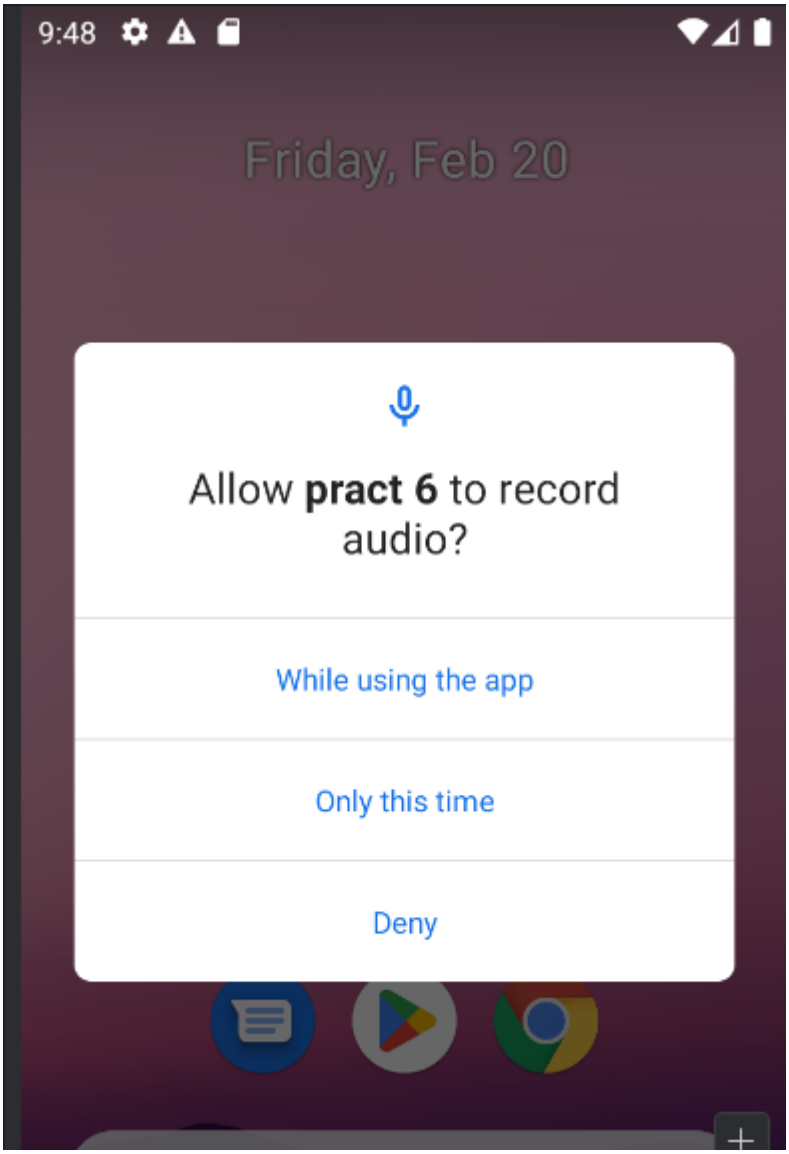
```
        fun onRequestPermissionsResult(
            requestCode: Int,
            permissions: Array<String>,
            grantResults: IntArray
        ) {
            //super.onRequestPermissionsResult(requestCode, permissions, grantResults, deviceId)
            when(requestCode)
            {
                REQUEST_RECORD_CODE->{
                    if (grantResults.isEmpty()||grantResults[0]!=
PackageManager.PERMISSION_GRANTED)
                    {
                        Log.d(TAG,"Permission denied")
                    }
                    else
                    {
                        Log.d(TAG,"Permission granted")
                    }
                }
            }
        }
    }
}
```

**AndoirdManifest.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.RECORD_AUDIO"></uses-
permission>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Pract6">
        <activity
            android:name=".MainActivity"
```

```
        android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
  </application>
</manifest>
```

**OUTPUT:**



# Practical No. 8:

AIM: Programming Media API and Telephone API

## CODE:

### activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <EditText
      android:id="@+id/edittextPhone"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:ems="10"
      android:textSize="25dp"
      android:text="Enter Number"
      android:inputType="phone"/>
   <Button
      android:id="@+id/phonecall"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginTop="200dp"
      android:text="Place call"/>
</LinearLayout>
```

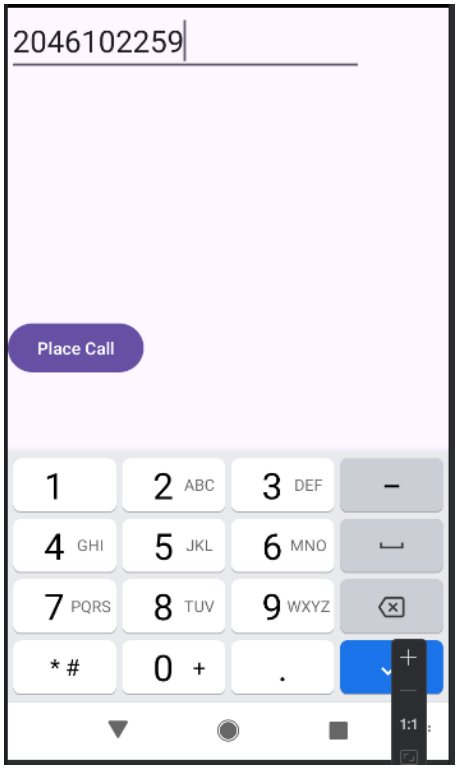
### MainActivity.kt:

```
package com.example.api
import android.content.Intent
import android.content.pm.PackageManager
import android.net.Uri
import android.os.Bundle
import android.os.PersistableBundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
```

```
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
class MainActivity : AppCompatActivity() {
   lateinit var btn: Button
   lateinit var txt: EditText
   override fun onCreate(savedInstanceState: Bundle?, persistentState: PersistableBundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)
      btn = findViewById<Button>(R.id.phonecall)
      txt = findViewById<EditText>(R.id.edittextPhone)
      btn.setOnClickListener {
         if
(ActivityCompat.checkSelfPermission(this,android.Manifest.permission.CALL_PHONE)!=
PackageManager.PERMISSION_GRANTED)
         {
            Toast.makeText(this,"Please grand permission",Toast.LENGTH_SHORT).show()
            requestPermissions()
         }
         else{
            val it = Intent(Intent.ACTION_DIAL)
            var number = txt.text
            it.data = Uri.parse("tel/$number")
            startActivity(it)
         }
      }
   }
   private fun requestPermissions()
   {
      ActivityCompat.requestPermissions(this,
arrayOf(android.Manifest.permission.CALL_PHONE),1)
   }
}
```

**AndroidManifest.xml:**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools">
<uses-permission android:name="android.permission.CALL_PHONE"></uses-permission>
   <application
      android:allowBackup="true"
```

```
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.API">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

**OUTPUT:**